



Equipment Placement & Wiring Guidelines

Guidelines for Fiber to the Unit Deployments of Spectrum Ready

Version 1.2 Updated: 1/30/25

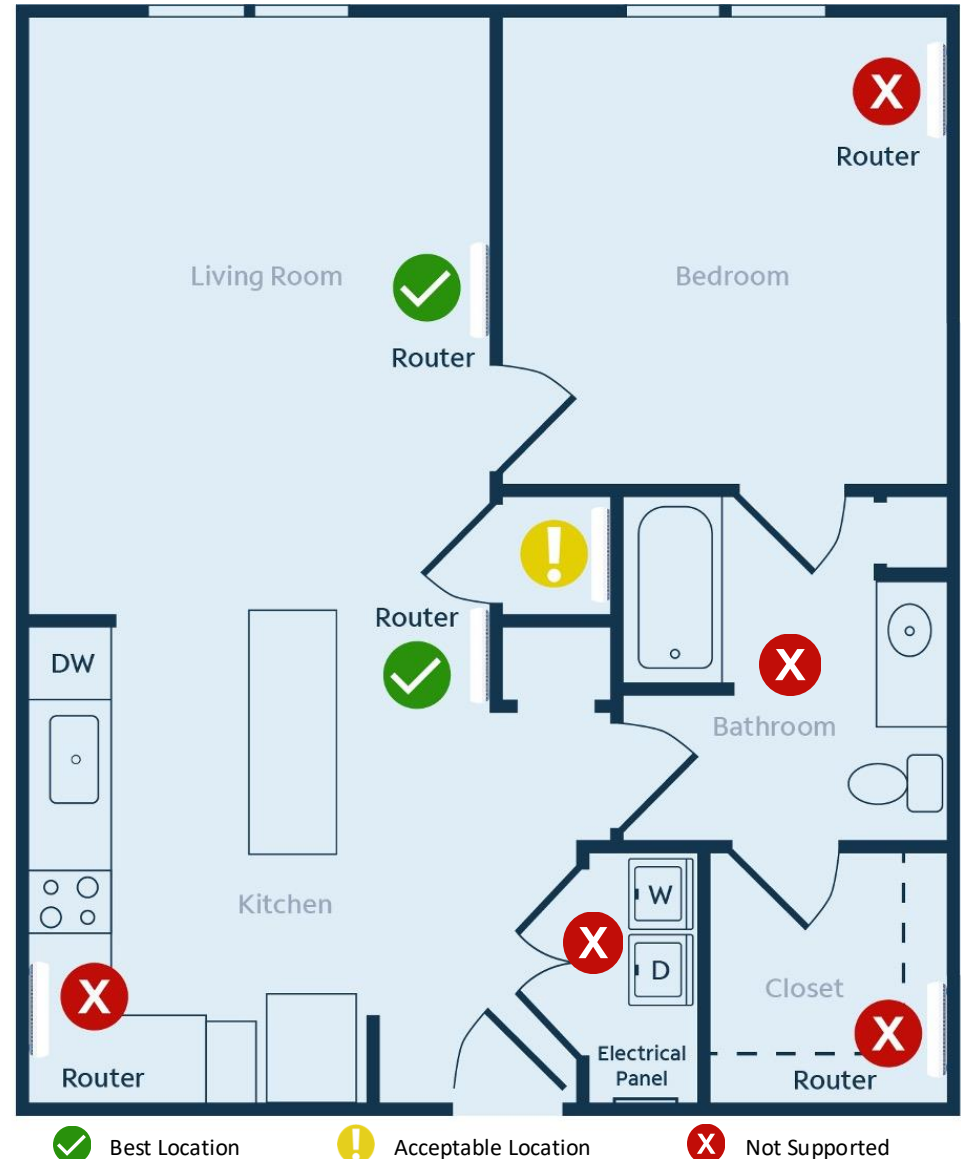
Where should I place the router in the floorplan?

IN A CENTRAL LOCATION THAT SUPPORTS OPTIMAL WIFI COVERAGE THROUGHOUT THE UNIT AND ALLOWS THE RESIDENT TO EASILY ACCESS THE ROUTER.

- Green locations are central to the apartment and are the best locations for the router to be placed for resident access and WiFi coverage.
 - Yellow locations are also central but may have reduced WiFi signal and resident access. These locations are acceptable, but not optimal. Closets are supported if central to the unit and if they do not contain appliances.
- Note:** If an acceptable closet contains the main electrical panel, the router should be placed away from the panel to avoid potential interference.
- Red locations are not supported as they do not provide optimal WiFi coverage. These include locations that are not central to the unit, as well as locations with appliances such as the kitchen, utility or laundry closets. Kitchen appliances, water heaters, and washer/dryers will degrade the WiFi signal. Additionally, bathroom mounting will not be supported.

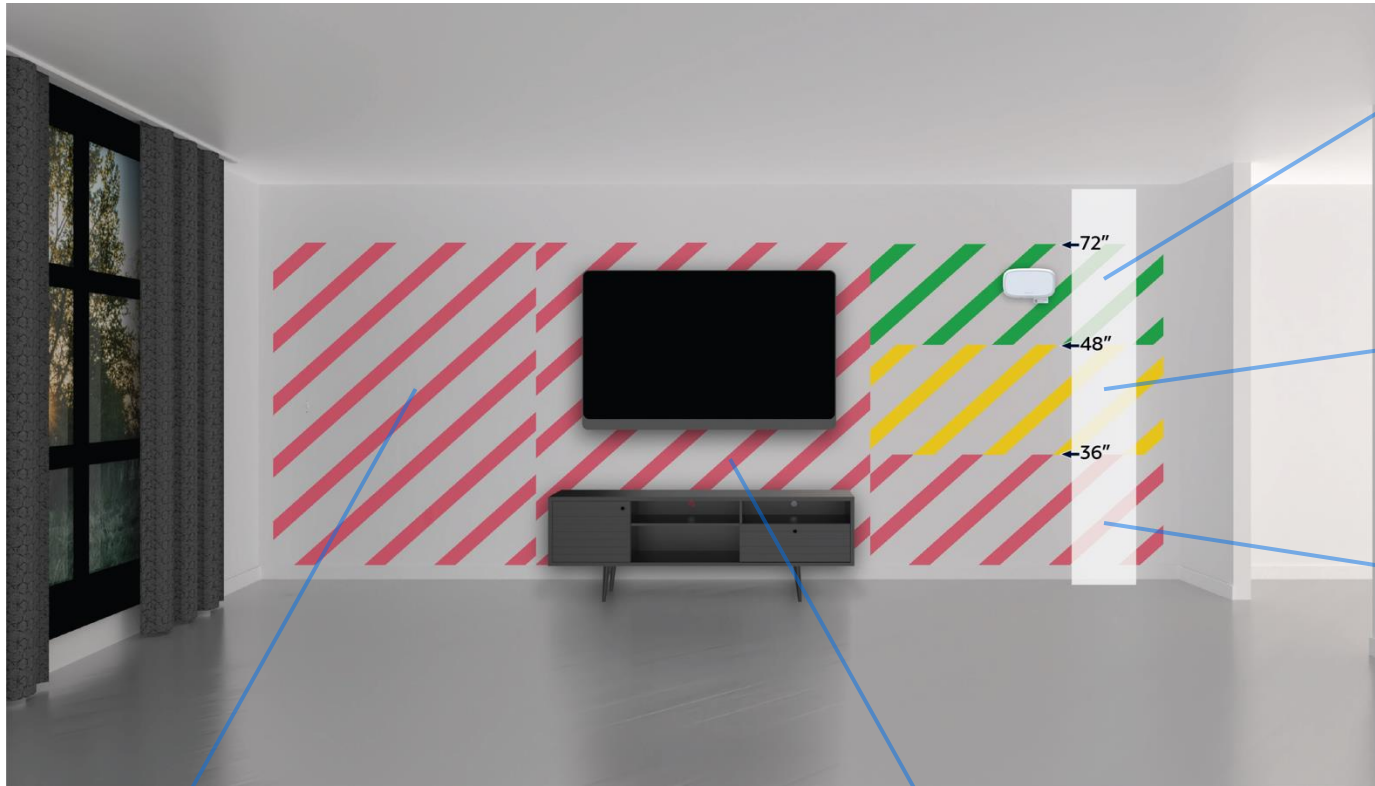
Use the following checklist when deciding where to locate the gang box and Cat6 for the router:

- ☐ Router is in a central location for WiFi coverage
- ☐ Router is away from appliances that cause WiFi signal degradation
- ☐ Resident must be able to easily see the router and scan the QR code
- ☐ Resident must be able to access the reset button and ethernet port on the bottom of the router



Where should the router go?

IN AN APPROVED SECTION OF A WALL IN A CENTRAL LOCATION OF A UNIT, ENSURING TO NOT BLOCK TV PLACEMENT OR POSITIONING IT TOO LOW ON THE WALL



Approved location, central to the home, at 48"-72" height away from the TV location

Approved location, central to the home at 36"-48" away from the TV location for ADA-compliant units

Installing the router below 36" will make it hard for residents to see the QR code, access the reset button, and use the ethernet port.

Router should not go here, as this side of the wall is not central to the unit

Router should never be installed centrally on the wall where a TV may go. Putting the router here could impede the TV being mounted on the wall or cause negative connectivity signal in the future.

How should the router be mounted on the wall?

Router positioned correctly on the wall with the QR Code, Reset Button, Ethernet port all accessible to the resident.



Router position is unacceptable as the QR Code, Reset Button, Ethernet port all sideways or upside down for the resident to access.



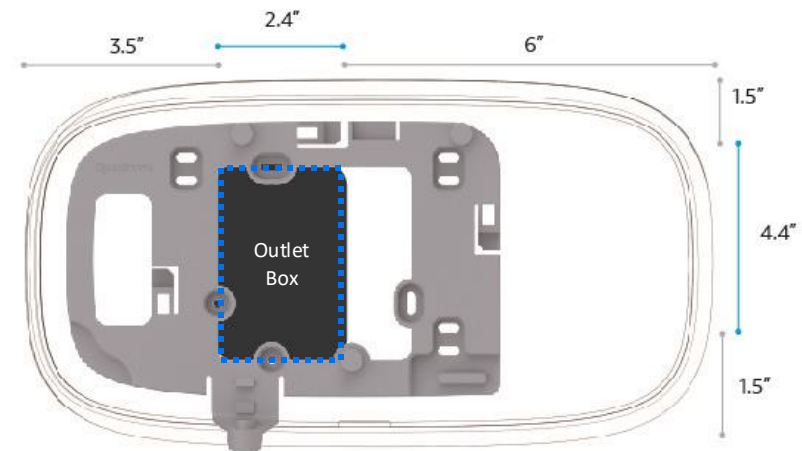
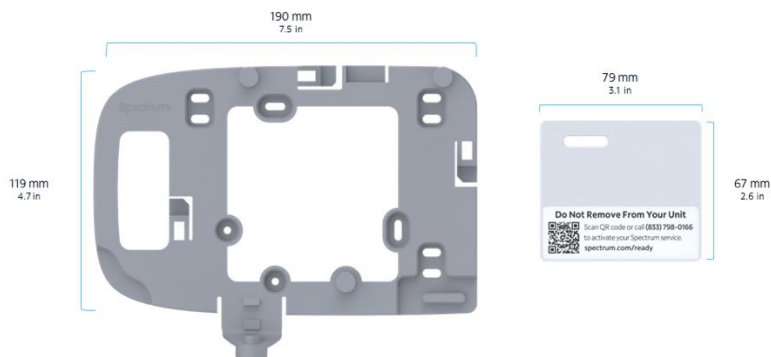
Installation Example with Wiring Installation



Equipment Dimensions



WiFi 6E MDU Router



Bracket Dimensions & Single Gang Outlet Mounting Clearance Requirements

Equipment Dimensions



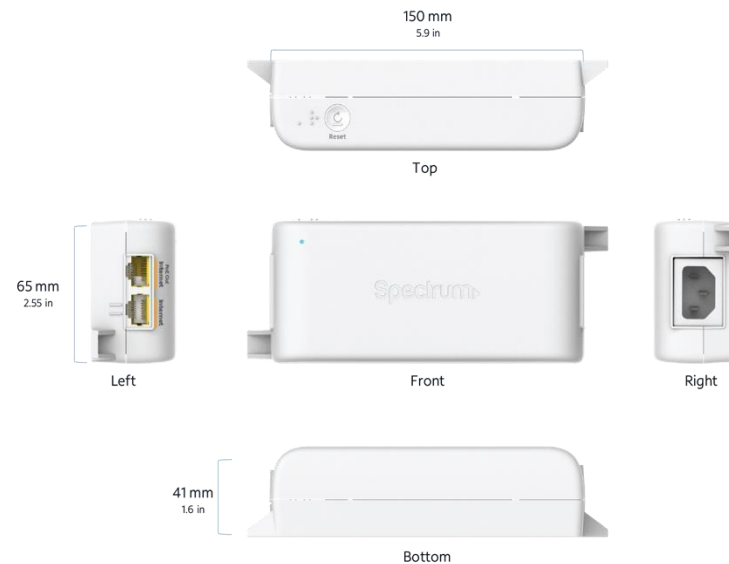
Media Panel
H: 30" W: 14" D: 4.7"

Primex P3000

S-ONU



Spectrum PoE Injector



Wiring Requirements

Requirements between Media Panel and MDF:

- Media Panel should be Primex P3000 or comparable panel in both size and material.
 - Inside Dimensions: H 30", W 14", D 4"
 - [Datasheet](#)
- Media Panel should contain one duplex NEMA 15A outlet.
- If an additional data switch is being provided in the media panel to support multiple hardwired ports, a second NEMA 15A outlet should be installed.
- Wiring from the MDF/IDF Telecom room must consist of an individual Fiber Optic Cable to each unit.
- Fiber Optic cable must be placed inside the Micro-Duct or conduit to the Media Panel.
 - Minimum Outside Diameter 10 mm Micro-Duct
 - Minimum Inside Diameter 8 mm Micro-Duct
- Supported Bend Radius 4"
- Fiber Minimum Specifications
 - Indoor/Outdoor Rated Single Mode Drop Fiber
 - Non-armored Construction Type
 - Gel-free
 - 900um tight-buffered.
 - 250-micron Fiber

Requirements for MDF:

- Developer to provide Main Distribution Frame (MDF) or Intermediate Distribution Frame (IDF) Telecom rooms.
- The MDF room will be the Minimum Point of Entry (MPOE).
- MDF room will distribute fiber to each IDF Telecom room.
- Minimum of (1) 2" conduit from exterior to MPOE MDF/IDF Telecom room.
- Minimum of (1) 3" conduit sleeve between stacked Telecom rooms on each floor.
- Vertical riser installs must be fire-sealed in accordance with the National Fire Protection Association.
- Minimum (1) 4' x 8' x ¾" fire-rated, plywood backboard in MPOE room marked "CATV".
- Common Ground at all specified MDF/IDF Telecom locations per NEC code based on approved design.

Requirements for wiring from Media Panel to Router

Location:

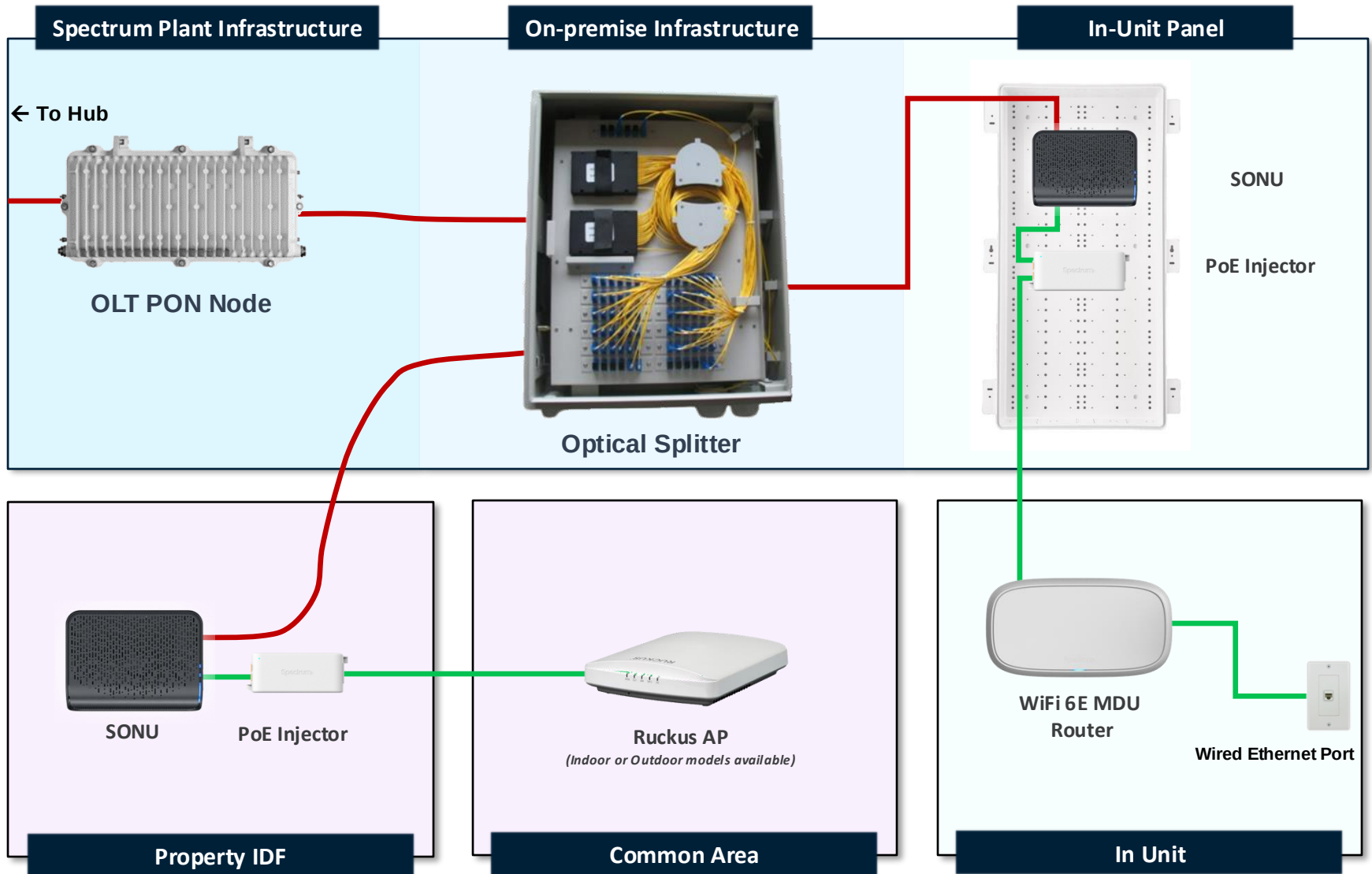
- Existing properties must use Category 6 (CAT6) as a minimum standard cable from the Media Panel to the router and data outlet(s) labeled at both ends and must conform to EIA/TIA 568-B wiring specifications.
- Greenfield (New Construction) must use Category 6 (CAT6) minimum standard cable from the Media Panel to the router and data outlet(s) labeled at both ends and must conform to EIA/TIA 568-B wiring specifications; for distances exceeding 180', Augmented Category 6 (CAT6a) is required.

Requirements for Router Location and Wired Ports:

- Dedicated, single Gang Junction Box with Ethernet cabling to support Router Placement.
- Greenfield Properties
 - The junction box for the router should be located on a centrally located wall that will be unobstructed, at a height between 48"-72" from the floor, to provide optimal Wi-Fi performance for the installed unit and any adjacent units.
 - Centralized location, ideally a common space (ex: Living Room)
- Placement of junction box to be marked clearly on Floor Plan with special instructions included regarding placement (if applicable).
- Physical identifier in each unit before installation to confirm desired placement during installation.
- CAT6 minimum standard cable from Media Panel to router location labeled at both ends. If the cable length is longer than 180' CAT6a must be used.
- If in-unit wired ports beyond router location:
 - Additional Category 6 CAT6 minimum standard cable from the Router location to the Media Panel must be installed and the cable must be labeled at both ends. If the cable length is longer than 180' CAT6a must be used.
 - Category 6 CAT6 minimum standard cable from the Media Panel to outlets must be installed and labeled at both ends. If the cable length is longer than 180' CAT6a must be used.
- If an additional data switch is being provided in the media panel to support multiple hardwired ports, a second NEMA 15A outlet should be installed.

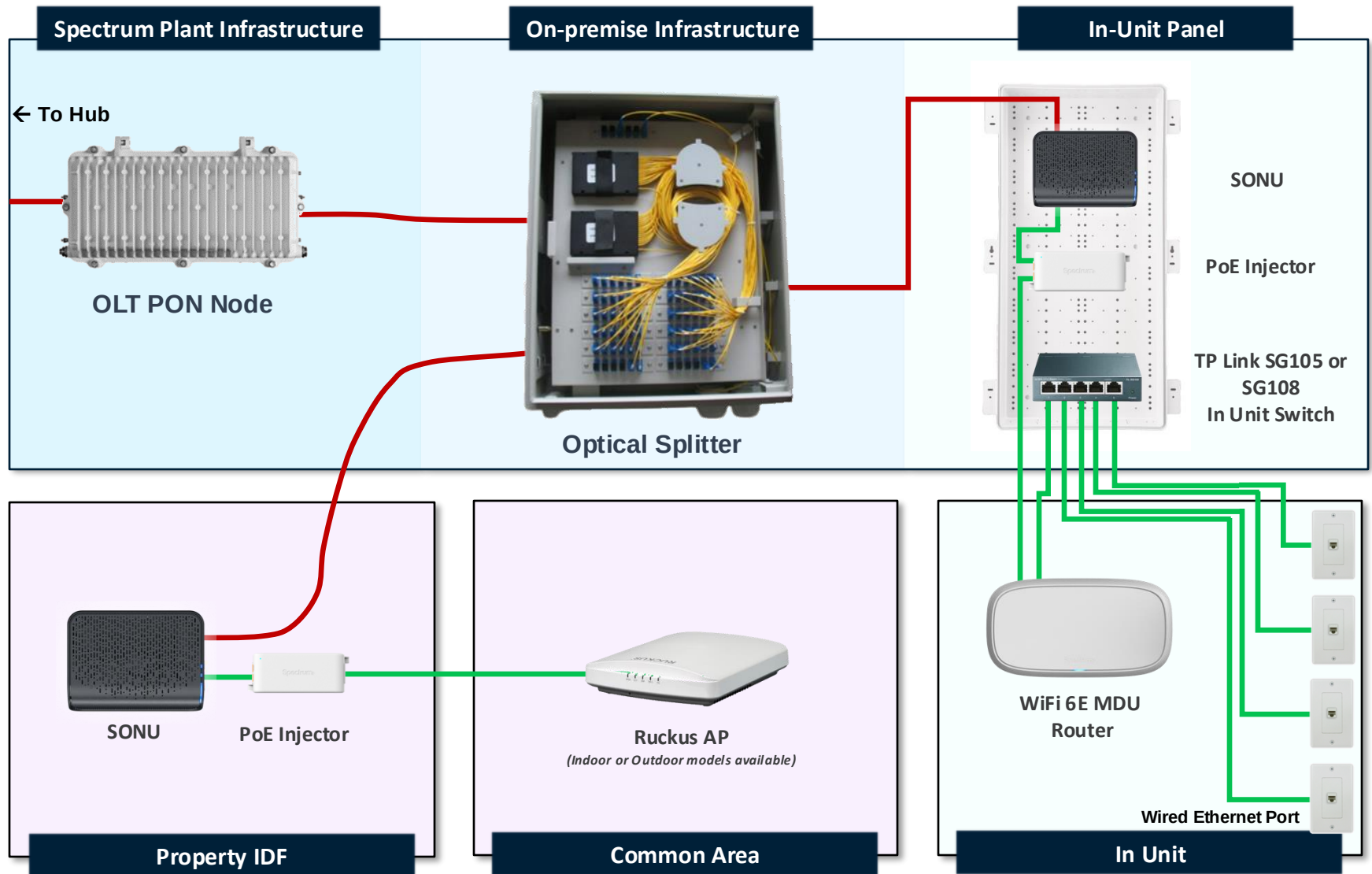
Spectrum Ready – Fiber to the Unit

WITH 1 WIRED PORT



Spectrum Ready – Fiber to the Unit

WITH 2-4 WIRED PORT



Notification to Owner/Developer

%Legal Entity%:

Included in this document are guidelines for in unit placement of Spectrum Ready routers and access equipment. Utilizing central locations that are easily accessible by residents is key to the performance and resident experience of this service. Please don't hesitate to ask your Account Executive or Service Delivery Project Manager (SDPM) if you have any questions or concerns regarding the router placement within the unit.

The internal wiring (as defined in the System Installation and Services Attachment) supporting the Spectrum Ready service at your property must adhere to the standards outlined by ANSI/TIA-568, including ethernet termination (ANSI/TIA-568-B). All installations must conform to the National Fire Protection Association (NFPA) and National Electrical Code (NEC) industry standards, as well as any state and local electrical codes.

The Developer / Owner is responsible for all permits required for the internal wiring and conduit placement, including microduct.

Operator is not responsible for WiFi coverage issues and may require owner to relocate wiring locations or equipment at their sole expense if router placement does not adhere to these guidelines.

Acknowledgement:

Signature: _____

Print Name: _____ %Owner Name% _____

Date: ____ / ____ / ____